

# Taha Rawjani

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## EDUCATION

**University of Pennsylvania** | B.S.E. Computer Science with EE Minor | 4.0 GPA

**Expected Spring 2029**

• Relevant Coursework: Multivariable Calculus, Statistics, Physics: Mechanics, Discrete Math, Intro to Algorithms, Game Theory

## EXPERIENCE

**Moby Labs** | Cofounder, CTO

**Jan 2026 – Present**

- Led the development of Keiko 1, an acoustic monitor for vessels, fishing boats, and offshore compliance supporting 300+ whale species.
- Designed novel ensemble CBAM architecture with a Bidirectional LSTM to detect and localize underwater whale calls. AUROC of 0.97
- Scraped data and labeled 30,000+ whale audio clips from Cornell University, the Watkins Marine Mammal Database, and NOAA.
- Modeled, assembled, and stress-tested (Von Mises) a solar-powered 3D-printed buoy in Fusion360. Printed with PETG in X1C.
- Designed a custom Hydrophone and preamp circuit, costing \$40 (2x cheaper than Aquarian) for whale calls. Used ESP32 for recording.
- Working with and demo-ing to 20+ whale watchers and NASA GSFC. Invited to YC Startup School and Telora Decision Day.

**Blue Vigil** | Embedded Software Engineer

**April 2026 – Aug 2026**

- Wrote embedded C++ for a tethered drone light system across Teensy, Raspberry Pi, 8-bit AVR, and Cube microcontrollers.
- Implemented a RS485-powered star comms network on ground and drone subsystems. Built custom ALEDMessage serialization protocol to transmit relayed messages and communicate state machine statuses through UDP, I2C, RS485, and UART devices.
- Refactored multi-board firmware monorepo into modular architecture, adding safety monitors, fault handlers, and CI/CD pipelines.
- Developed an open-source Over-The-Air (ota) remote update software and bootloader for Teensy4.1 and ATmega boards.
- Designed and fabricated a custom AVR-powered spool controller PCB in KiCad and wrote bootstrap firmware for 4+ similar AVR boards.
- Led codebase migration from C++ to embedded rust, and from Raspbian to NixOS for safety and better consistency.

**Maximus** | Software Engineering Intern

**Jun 2025 – Aug 2025**

- Created an AWS pipeline to classify email intents, reroute emails, and autonomously respond with AI – reducing agent resolve times by 25%. Fully automatable and customizable through JSON w/ GUI configuration tool.
- Used AWS Connect, Lambda, Bedrock, EC2, Cloudwatch, Kinesis, alongside frontier AI models for reasoning.
- Pitched to Maximus CTO, handed off and documented project to Contact Center SWE team for deployment.

## AWARDS & HONORS

**Competitive Programming, Hackathons, and CTFs**

- Codeforces Candidate Master and USACO Platinum Contestant. 1500+ problems solved across CF, USACO, LeetCode, CSES.
- Team USA for International Informatics Olympiad in Teams (IIOT) 2025, 2nd Place (out of 500+) USIOT 2025
- 1st place (out of 6000+) in Category, 4th in USA, National Cyber League Team Round Fall 2025; 1st place CCI Meta CTF 2025.
- 1st place Advanced Division at Lockheed Martin CodeQuest 2024 and 2026; 3rd at VCU HSPC 2025 and 2026
- 1st place at UPenn PClassic 2024; 2nd place at VT HSPC 2024; 3rd at UVA HSPC 2026; 2nd at UMD HPSC 2025
- 1st place 2023, Best Game 2024, and 2nd place 2025 at ACLHacks; Education track winner PatriotHacks; 1st place PhoenixHacks 2024

**Research Awards**

- International Science and Engineering Fair (ISEF) 2025 4th place grand & special award at in Robotics & Intelligent Machines.
- Recognized in Congressional Record (H1227) by Rep. Suhas Subramanyam for Whale Conservation Research.
- 1st place in Category, Special, and Grand Award at LCPS RSEF 2025. 3rd place in Category at VSSEF 2025.

## PUBLICATIONS & PATENTS

**T. Rawjani, M. Li.** “REAL-TIME MARINE MAMMAL IDENTIFICATION AND LOCALIZATION.”: U.S. Patent Application No. 19/720,276, filed 2026. Patent pending.

**T. Rawjani, M. Li.** “MobyGlobal: Real-Time Whale Detection Network Powered by a Two-Branch Ensemble Learning Model on 3D-Printed Buoys.” *Ocean Sciences Meeting 2026*, February 2026, Glasgow, Scotland. E-Lightning.

**T. Rawjani, M. Li.** “MobyGlobal: A real-time whale tracking network” *Intl. Conference on Marine Science and Eng (IOCMSE)* 2025

## PROJECTS

**Neo OS**

**Jul. 2022 - Aug. 2023**

- An operating system written from scratch in C++ and assembly with a UNIX-based virtual file system, and multithreading support.
- Wrote TahaScript, a dynamically interpreted scripting language; and TWM, a customizable window manager built for Neo-OS.
- Programmed an in-house image format library (.nic) with 24-bit color support alongside custom font and VGA drivers
- Created drivers for hard drives (AHCI), floppy disks (FDC), timers (PIT, APIC), interfaces (PS/2 mouse/kbd, PCI), filesystems (FAT32)

**6502 GPT**

**June 2026 - July 2026**

- Built a Ben-Eater inspired 6502 breadboard computer using a 8-bit W65C02S MPU, W65C22S Interface adaptor, AT28C256 32K EEPROM, AS6C62256 RAM, and a custom adjustable 555-based clock (65 kHz).
- Programmed an On-Chip debugger tracing 6502 clock, stack, registers, interrupts, and instructions using an Arduino Mega.
- Wrote an OS with vasm-oldstyle and cc65 cross compilers, supporting I2C and serial communication with LCDs, SD storage, and keypads.
- Ported custom C-based deep learning library (CMLIB) and ran an in-house small-paramter LLM on 4 MHz clock speed.

## TECHNICAL SKILLS

**Electrical & Embedded:** Embedded C/C++, Microcontrollers (Arduino, ESP32, Teensy4.1), MPUs (6502, AVR), Raspberry Pi, HV power systems and rectifiers, multi-board firmware architecture, Fusion 360, 3D printing, KiCad PCB Design, Circuit Analysis

**Software Engineering & Programming:** C++, C, Python, Rust, TypeScript/JavaScript, x86 & 6502 assembly, Git, AWS (Lambda, Bedrock, EC2, Connect, Kinesis, CloudWatch), Supabase, Next.js, REST APIs, web scraping, operating systems (device drivers, virtual file systems, multithreading), protocol design, language & interpreter design

**Machine Learning:** PyTorch, TensorFlow, ensemble learning, attention mechanisms, acoustic signal classification, intent classification, LLM development, agent orchestration (LangGraph, MCP)